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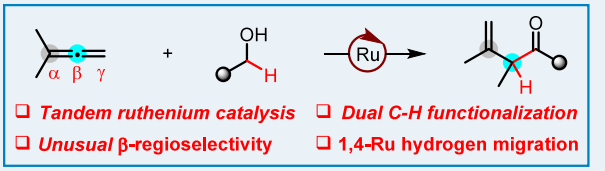
Ruthenium-Catalyzed Regioselective Migratory Hydroacylation of Nonactivated Allenes with Alcohols Enabled by 1,4-Ru/H Migration

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ABSTRACT: 1,4-Metal/hydrogen migration is a flexible strategy that not only activates remote C−H bonds but also mediates regioselective coupling reactions. We describe here an intermolecular coupling between allenes and alcohols to access β,γ-unsaturated ketones via tandem ruthenium catalysis. The reaction occurs via migratory hydroacylation of allenes with an in situ-generated aldehyde through the dehydrogenation of the alcohols. This formal dual C−H functionalization reaction enables one-step efficient synthesis of β,γ-unsaturated ketones from nonchelating alcohols using a commercially available catalyst and reagent. Unusual β-regioselectivity has been achieved with C−C coupling at the central *sp*-hybridized carbon of allenes. Moreover, β-regioselective migratory hydroacylation of allenes with nonchelating aldehydes has also been disclosed. The detailed mechanism has been proposed based on the control experimental and computational studies involving hydrogen transfer catalysis and key 1,4-ruthenium hydrogen migration.



KEYWORDS: *C−H functionalization, 1, 4-Ru/H migration, hydroacylation, allene, alcohol*

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INTRODUCTION

Metal-catalyzed C−H functionalization plays a significant role in many fields of chemical science.[1−6] Among them, C−H functionalization by 1,4-metal/hydrogen migration has recently attracted considerable attention in various transition-metal-catalyzed coupling reactions. The migration process of 1,4-metal/hydrogen is not only mechanistically fascinating but also a key step to realize novel transformations.[7−9] To date, tremendous reactions involving the 1,4-metal/hydrogen migration mediated by palladium,[10−12] rhodium,[13,14] nickel,[15−20] platinum,[11] cobalt,[21,22] ruthenium,[23,24] iridium,[25] iron,[26] chromium,[27] and copper[28] have been achieved (Figure 1A). The general mechanism for these 1,4-metal/hydrogen migration follows an oxidative insertion to generate a five-membered cyclometalated intermediate and then a reductive elimination pathway, enabling the migration of diverse C−H groups, including aryl, vinyl, benzyl, and alkyl moieties. Allenes have received extensive interest as versatile synthetic building blocks because of their unique features since Van't Hoff's original predictions about structure and composition.[29,30] Recently, Ma and co-workers reported a novel and efficient 1,4-hydrogen delivery process mediated by allylic rhodium species to synthesize α,β-unsaturated ketones from α-allenyl-substituted alcohols (Figure 1B).[31] Base-mediated dehydrogenation of alcohol to ketone is the driving force of 1,4-hydrogen delivery in this reaction. Despite these significant advances, the application of 1,4-metal/hydrogen migration in useful transformations with allenes has rarely been reported. To further exploit the potential of 1,4-metal/hydrogen migration in the hydrofunctionalization of allenes, a new strategy of 1,4-metal/hydrogen migration is highly desirable.

Different from alkene and alkyne, allene exhibits three reaction sites coupled to a potential axial chirality. It is fascinating and challenging to achieve the site- or regioselective transformations of allenes.[32−37] In particular, the hydrofunctionalization of allene is a convenient and useful method to construct functionalized alkenes, which usually occurs at the α- or γ-position of allene due to its structural nature and special bonding properties.[38] Metal-catalyzed cross-coupling reactions between allenes and alcohols have been reported. The efficient Pd-, Rh-, or Au-catalyzed hydroalkoxylation of allenes with alcohols has been reported to synthesize allylic ethers via a Pd/Rh−H insertion/π-allyl-Pd/Rh formation process or a cationic Au-allene intermediate.[39−41] Elegant Ir- or Ru-catalyzed hydrohydroxyalkylation of allene with alcohols has been reported, affording homoallylic alcohols with exclusive α-regioselectivity by Krische and others with an in situ-generated aldehyde (Figure 1C).[42−47] This strategy of using alcohol as both a reductant and carbonyl pro-electrophile has been widely used to enable the conversion of alcohols to various useful products.[48−52] We envisioned that the regioselective insertion of a metal hydride (M−H) species into an alkyl-substituted allene

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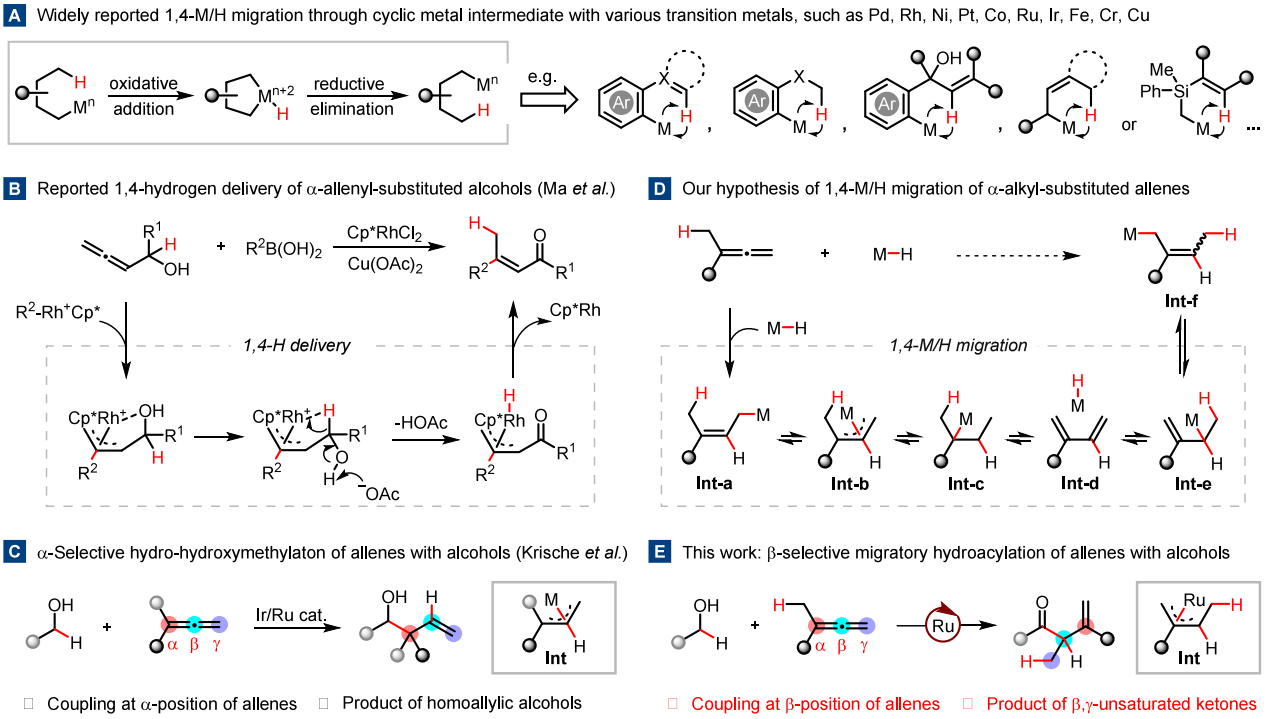
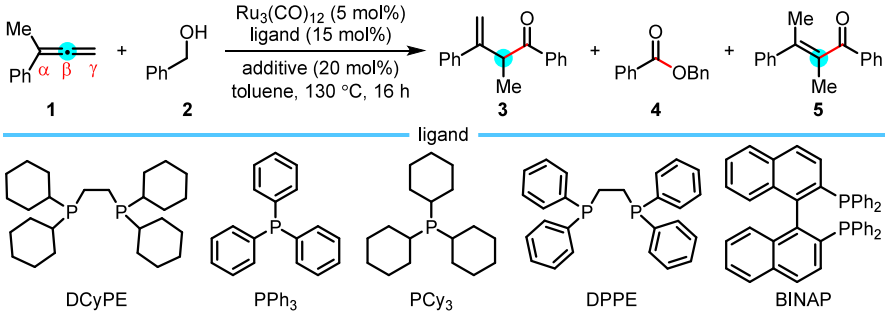


Figure 1. 1,4-M/H migration and regioselective hydrofunctionalization of allenes.

Table 1. Effects of the Reaction Parameters on the Reaction Efficiency[a]



Entry	Ligand	Additive	Yield of product (%)[b] 3	4	5
1	DCyPE	-	30	30	0
2	PPh_3	-	<5	0	0
3	PCy_3	-	5	0	0
4	DPPE	-	0	0	0
5	BINAP	-	10	0	0
6	DCyPE	K_3PO_4	0	0	0
7	DCyPE	AgOTf	42	0	31
8	DCyPE	$NaBF_4$	62	0	12
9	DCyPE	$(PhO)_2PO_2H$	70	0	0
10[c,d]	DCyPE	$(PhO)_2PO_2H$	90 (84)	0	0

[a]Reaction conditions: **1** (0.4 mmol), **2** (0.2 mmol), $Ru_3(CO)_{12}$ (5 mol %), ligand (15 mol %), additive (20 mol %), toluene (2 mL), 130 °C, and 16 h, unless otherwise noted. [b]Determined by crude ^1H NMR with 1,1,2,2-tetrachloroethane as an internal standard. [c]With **1** (0.5 mmol), DCyPE (20 mol %), 24 h, and isolated yield in parentheses. [d]Gram-scale reaction with **1** (20 mmol) and **2** (8 mmol). **3** was obtained in 66% yield.

at its less bulky double bond would generate allyl metal intermediate **Int-a** (Figure 1D). Tautomerization via π-allyl-metal intermediate **Int-b** forms allyl metal intermediate **Int-c**. Next, β-hydrogen elimination generates diene **Int-d** and the M−H species. Selective reinsertion of M−H species into the diene affords allyl-metal intermediate **Int-e**. Then, allylic tautomerization again generates allyl-metal intermediate **Int-f**. Thus, a process of 1,4-metal/hydrogen migration is realized. If the M−H species is generated via metal-mediated dehydrogenation of an alcohol, then an aldehyde should be formed simultaneously. The following coupling of an aldehyde with intermediate **Int-e** or **Int-f** will lead to the corresponding products with different chemo- and regioselectivities. The challenge is to find a suitable catalytic system to achieve the selective cross-coupling of allene

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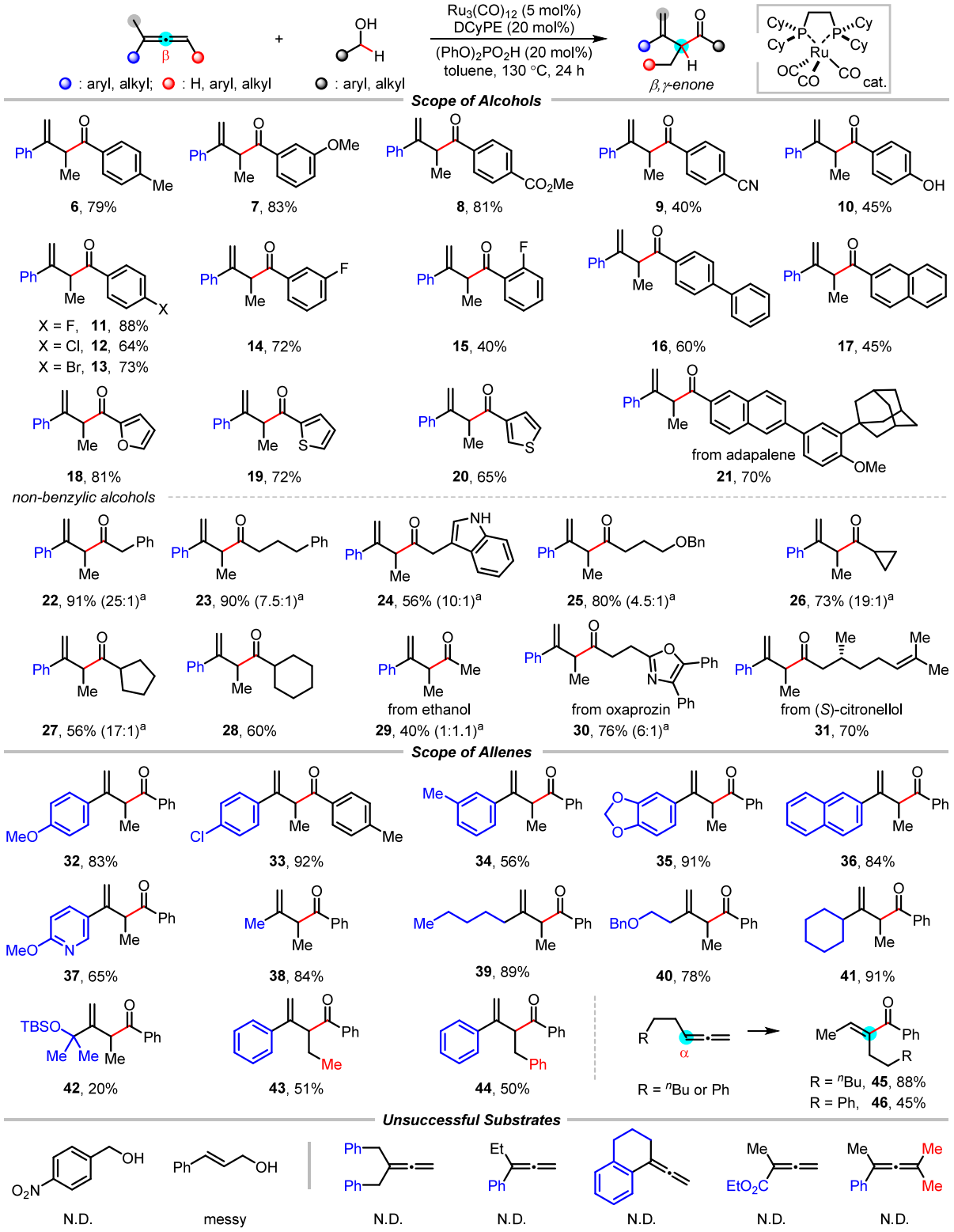


Figure 2. Reaction scope of migratory hydroacylation of allenes with alcohols. Reaction conditions: alcohol (0.2 mmol), allene (0.5 mmol), $Ru_3(CO)_{12}$ (0.01 mmol), DCyPE (0.04 mmol), diphenyl phosphate (0.04 mmol), toluene (2.0 mL), 130 °C, and 24 h. The isolated yield of product is provided.

and alcohol. Here, we report the first migratory hydroacylation of allenes with alcohols to produce useful β,γ-unsaturated ketones, which gives the ketone product with exclusive β-regioselectivity enabled by hydrogen-transfer tandem Ru catalysis (Figure 1E).

METHODS

General Procedure for the Ru-Catalyzed Hydroacylation of Allenes with Alcohols. To a nitrogen-filled, oven-dried 25 mL Schlenk tube containing a stirrer bar was added allene (0.5 mmol), alcohol (0.2 mmol), $Ru_3(CO)_{12}$ (6.4

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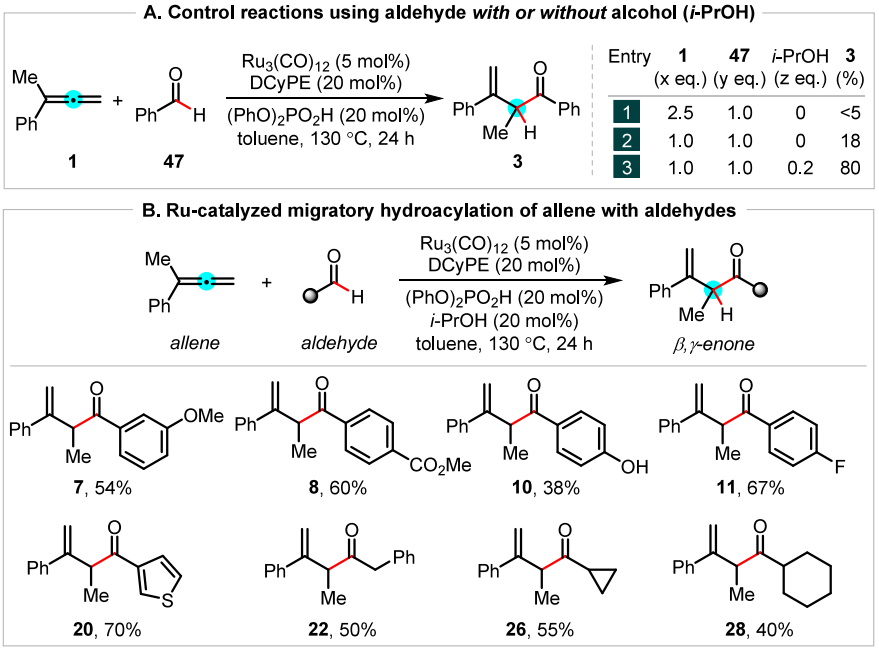


Figure 3. Investigation of Ru-catalyzed migratory hydroacylation with aldehydes.

mg, 0.01 mmol), 1,2-bis(dicyclohexylphosphino)ethane (DCyPE, 17.0 mg, 0.04 mmol), $(PhO)_2PO_2H$ (10.0 mg, 0.04 mmol), and dry toluene (2.0 mL). The Schlenk tube was sealed, taken out of the glovebox, and stirred at 130 °C for 24 h. Then the reaction mixture was cooled to room temperature, concentrated under reduced pressure, and purified by flash column chromatography (eluent: petroleum ether/ethyl acetate) to give the desired enone product.

General Procedure for the Ru-Catalyzed Hydroacylation of Allenes with Aldehydes. To a nitrogen-filled, oven-dried 25 mL Schlenk tube containing a stirrer bar was added allene (0.2 mmol), aldehyde (0.2 mmol), $Ru_3(CO)_{12}$ (6.4 mg, 0.01 mmol), 1,2-bis(dicyclohexylphosphino)ethane (DCyPE, 17.0 mg, 0.04 mmol), $(PhO)_2PO_2H$ (10.0 mg, 0.04 mmol), ^{i}PrOH (3 μL, 0.04 mmol), and dry toluene (2.0 mL). The Schlenk tube was sealed, taken out of the glovebox, and stirred at 130 °C for 24 h. Then the reaction mixture was cooled to room temperature, concentrated under reduced pressure, and purified by flash column chromatography (eluent: petroleum ether/ethyl acetate) to give the desired enone product.

RESULTS AND DISCUSSION

We began our investigation by studying the cross-coupling of simple 1-methyl-1-phenylallene (**1**) with benzyl alcohol (**2**). A variety of metal and ligand combinations were employed to catalyze the reaction. After an enormous number of investigations, we were glad to find that ketone **3** was obtained in 30% yield when commercially available $Ru_3(CO)_{12}$ was used as the catalyst and 1,2-bis(dicyclohexylphosphanyl)ethane (DCyPE) was used as the ligand in refluxed toluene (Table 1, entry 1). Notably, coupling at the central *sp*-hybridized carbon of allene occurred, exclusively leading to the β-selective migratory hydroacylation product. Neither an α- nor γ-selective hydroacylation nor hydrohydroxyalkylation product was detected. Benzyl benzoate (**4**) was also obtained as a byproduct in 30% yield under these conditions through the Ru-catalyzed homocoupling of benzyl alcohol.[53,54] A series of other ligands and Ru catalysts were tested, which showed lower or no reactivity, such as PPh_3, PCy_3, DPPE, BINAP, $RuCl_3$, $RuHCl(CO)(PPh_3)_3$, and $RuH_2(CO)(PPh_3)_3$ (Table 1, entries 2−5, and Table S1 in the Supporting Information). Then, a variety of additives were investigated to improve the efficiency of the migratory hydroacylation reaction. No reaction occurred when an inorganic base (K_3PO_4) was used as the additive (Table 1, entry 6). The yield of **3** was increased with AgOTf or $NaBF_4$ as an additive, but the isomerized product of α,β-unsaturated ketone **5** was also observed (Table 1, entries 7 and 8). Encouragingly, a clean reaction and a good yield of **3** were obtained in the presence of a catalytic amount of diphenyl phosphate (Table 1, entry 9). Finally, the best result was obtained with the isolation of **3** in 84% yield when the reaction time was prolonged and the dosage of reactants was slightly adjusted (Table 1, entry 10).

Having identified the optimal coupling conditions, the scope of the alcohols was first investigated with the coupling of allene **1**, and representative examples have been summarized in Figure 2. In general, benzylic and nonbenzylic alcohols both worked smoothly, giving the corresponding β,γ-unsaturated ketones in good to excellent yields with exclusive β-regioselectivity. The reaction tolerates various functional groups at the para-, meta-, and ortho-positions on the phenyl ring in the benzyl alcohols, such as electron-donating Me and OMe (**6** and **7**), electron-withdrawing ester and cyano (**8** and **9**), free hydroxy (OH, **10**), and halide (F, Cl, Br, **11**−**15**) groups. Alcohols bearing biphenyl (**16**), naphthalene (**17**), and heteroarenes (thiophene, furan, indole, and oxazole) were also compatible, providing the desired products of unconjugated enones in good yields (**18**−**20**, **24**, and **30**). The desired products were obtained in moderate to good yields with various nonbenzylic alcohols (**22**−**31**). And partial isomerization of β,γ-unsaturated ketones to α,β-unsaturated ketones was observed with the coupling of nonbenzylic alcohols. Of particular note is using ethanol as the substrate, which can be successfully converted to the desired ketone product (**29**). Alcohols derived from drugs (adapalene and oxaprozin) or a natural product (citronellol) also run smoothly, providing the desired targets (**21**, **30**, and **31**),

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presaging the utility of this method in the late-stage functionalization of complex molecules. No reaction occurred with 4-nitro benzyl alcohol, and a messy reaction was observed with cinnamyl alcohol.

Next, the scope of allenes was explored. Generally, 1,1-disubstituted and trisubstituted allenes containing an α-methyl group all work smoothly, providing the corresponding β-regioselective migratory hydroacylation product. The reaction tolerates a serious of functional groups on the aromatic ring of 1-methyl-1-arylallenes, including functionalized aryl, naphthyl, and pyridyl groups (**32**−**37**). In addition, unactivated 1-methyl-1-alkylallenes also reacted smoothly, affording the corresponding products in 78−91% yields (**38**−**41**), including the simple 1,1-dimethylallene (**38**). The coupling of allene with an α-substituted bulky group also delivered desired product **42** in a promising yield. Moreover, trisubstituted allenes were also proven amenable, generating β,γ-unsaturated ketones **43** and **44** in good yields. Interestingly, monosubstituted allenes showed different reactivities under the standard conditions. The α-regioselective hydroacylation of 1-hexylallene or 1-phenylethylallene was achieved, affording conjugated enones in moderate to good yields (**45** and **46**). The corresponding product was not detected with allenes without the α-methyl group, allenyl ester, and tetrasubstituted allenes.

To monitor the process of this migratory hydroacylation of allene **1** with benzyl alcohol (**2**), we observed the benzaldehyde (**47**) by thin layer chromatography and GC analysis under standard conditions. Aldehyde **47** disappeared after the reaction. These results indicate that the aldehyde should be the reaction intermediate, which is generated by Ru-mediated dehydrogenation of benzyl alcohol. However, a trace amount of hydroacylation product **3** was observed when directly replacing benzyl alcohol (**2**) with benzaldehyde (**47**) under standard conditions. And a small amount of corresponding alkenes derived from allene **1** was detected (Figure 3A-1). When the ratio of allene **1** and aldehyde **47** was reduced to 1:1, **3** was obtained in 18% yield (Figure 3A-2). Furthermore, when 20 mol % isopropanol was used as an additive, the migratory hydroacylation with aldehyde **47** took place efficiently, producing the desired product **3** in 80% yield (Figure 3A-3). These results provide the following clues for the reaction: (1) In situ-generated aldehyde through the dehydrogenation of alcohol is the reaction intermediate. (2) The corresponding Ru-hydride species should be the active catalyst, which can be generated by reaction with both $(PhO)_2PO_2H$ and alcohol. (3) The presence of alcohol can dramatically increase the hydroacylation reaction.

Then, the scope of the β-regioselective migratory hydroacylation of allene with aldehydes was also simply investigated (Figure 3B). The reaction occurred smoothly with a 1:1 ratio of allene and aldehyde in the presence of a catalytic amount of isopropanol under otherwise the same conditions. Aromatic and heteroaromatic aldehydes gave the corresponding β,γ-unsaturated ketones in comparable or lower yields compared to the benzylic alcohol substrates (**7**, **8**, **10**, **11**, and **20**). When alkyl aldehydes were used as the hydroacylation reagents, the desired products were obtained in lower yields compared to the utilization of the corresponding alcohols (**22**, **26**, and **28**). Using this strategy, a regioselective atom-economic migratory hydroacylation is developed.

The β,γ-unsaturated ketones are versatile synthetic intermediates since both ketone and olefin double bonds can be converted to various functional groups. As shown in Figure 4, several synthetic transformations have been carried out to

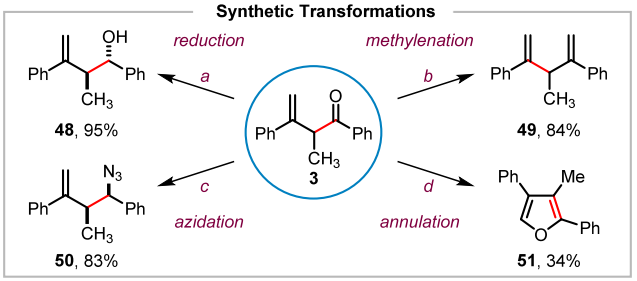


Figure 4. Synthetic transformations of the β,γ-unsaturated ketone product.

demonstrate the synthetic utilities of the β,γ-enone product. Stereoselective reduction of unconjugated enone **3** with $NaBH_4$ afforded antitype homoallylic alcohol **48** in 95% yield. Next, Wittig olefination of **3** with phosphorus ylide generated 1,4-diene **49** in a high yield. Furthermore, sequential reduction of **3** and the Mitsunobu reaction with phosphoryl azide produced syn-type homoallylic azide **50** in 83% yield. Finally, trisubstituted furan **51** was obtained via epoxidation of the double bond in enone **3** followed by acid-catalyzed cyclization.

To deeply understand the mechanism of this Ru-catalyzed β-selective migratory hydroacylation of allenes, a variety of control experiments were further conducted. Since an excess amount of allene is used as the substrate, we first tried to determine the byproducts of allene. To facilitate the isolation and analysis, we selected allene **52** as the substrate. Besides the desired migratory hydroacylation product **36**, internal alkenes *E*-**53** and *Z*-**53** were obtained as the major byproducts along with a small amount of terminal alkenes via the regioselective hydrogenation of allene (Figure 5A). This indicates that allene is both a coupling partner and a hydrogen sacrificer. Then, deuterium labeling experiments were performed with deuterated alcohol or allene under the standard conditions. First, migratory hydroacylation of allene **52** with deuterated benzyl alcohol d_2-**2** provided the product *d*-**36** in good yield, with deuterium (71% D at the α-position of ketone) majorly transferring to the β-carbon of allene (Figure 5B). In this reaction, deuterated alkene *d*-*E*-**53** was obtained as the major byproduct, with deuterium partially incorporating at the β- and γ-carbons of allene (14% D and 30% D, respectively). Interestingly, the minor byproduct alkene *d*-*Z*-**53** was obtained with significant deuterium (25% D) incorporation at the α-methyl group of allene. Then, migratory hydroacylation of deuterated allene d_3-**52** with alcohol **2** afforded the product *d*-**36** in good yield, with the transfer of deuterium (91% D) from the α-methyl of allene to the γ-carbon of allene (Figure 5C). Similarly, alkene byproducts *d*-*E*-**53** and *d*-*Z*-**53** were also obtained with similar deuterium transfer. These results indicate a clear process of 1,4-hydrogen migration, and the reaction should be a rapid and reversible process. Together with the conclusion in Figure 3A, deuterated benzaldehyde (PhCDO) is generated with alcohol d_2-**2** as the substrate during the reaction. Obviously, the carbonyl sp^2-hydrogen and acyl group of the aldehyde are both connected to the central *sp*-hybridized β-carbon of allene after the reaction. It is confirmed that the reaction does not proceed through a conventional acyl-Ru-hydride intermediate followed by migratory insertion into one double bond of allene.[55] Next, a crossover experiment with both deuterated and nondeuterated allene was carried out, and no crossover product was obtained (Figure 5D). This result suggests that the reaction occurs without the dissociation of ruthenium from the intermediates at any intervening stage. To

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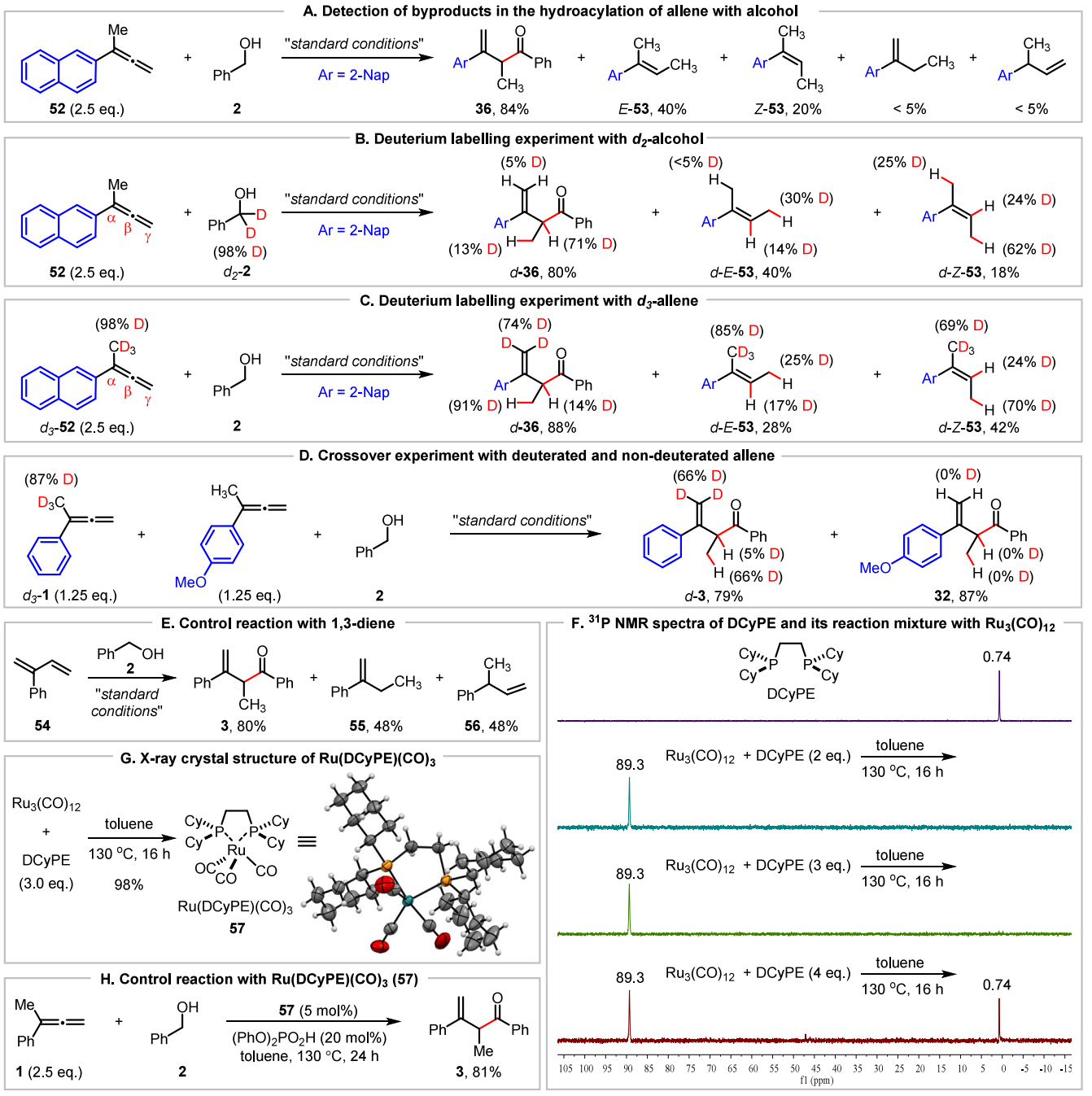


Figure 5. Control experiments and mechanistic studies.

further confirm the conjecture, the control reaction with 2-phenyl-1,3-butadiene (**54**) and alcohol **2** was also carried out under the standard conditions, and desired product **3** was obtained in 80% yield (Figure 5E). And terminal alkenes **55** and **56** were obtained simultaneously. This result suggested that an isomerization of allene to the 1,3-diene intermediate might take place during the reaction.

To further explore the coordination geometry of ruthenium with the diphosphine ligand DCyPE, we performed the following studies. First, the reaction of $Ru_3(CO)_{12}$ with 2 or 3 equiv of DCyPE (^{31}P NMR: 0.74 ppm) afforded the same DCyPE-coordinated Ru complex showing the identical single signal of ^{31}P NMR at 89.3 ppm (Figure 5F). When increasing the ratio of $Ru_3(CO)_{12}$/DCyPE to 1:4, besides the new signal, the ^{31}P NMR signal of residual DCyPE showed up. These results clearly confirm that the DCyPE-coordinated Ru complex was generated as a symmetrical compound with a 1:1 ratio of DCyPE to Ru. And both phosphine atoms of DCyPE coordinate to the Ru center because only one ^{31}P NMR signal is present in this complex. However, we do not know how many monooxides are coordinated to the Ru center in this stage. To our delight, this Ru complex was isolated in 98% yield when the reaction of $Ru_3(CO)_{12}$ with 3 equiv of DCyPE was performed in refluxed toluene (Figure 5G). Its structure was further confirmed as $Ru(DCyPE)(CO)_3$ (**57**) by X-ray crystallography analysis. Furthermore, a control experiment was carried out to verify the catalytic activity of $Ru(DCyPE)(CO)_3$ (**57**) in our β-regioselective migratory hydroacylation of allene **1** with alcohol

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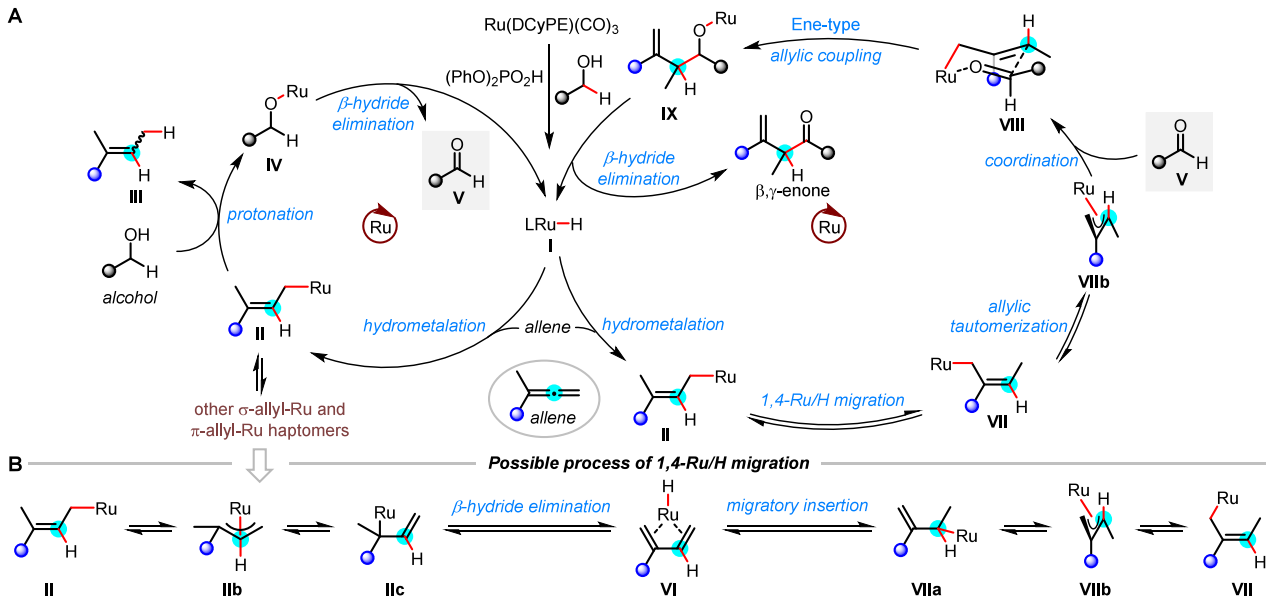


Figure 6. A. Plausible reaction mechanism. B. Possible process of 1,4-Ru/H migration.

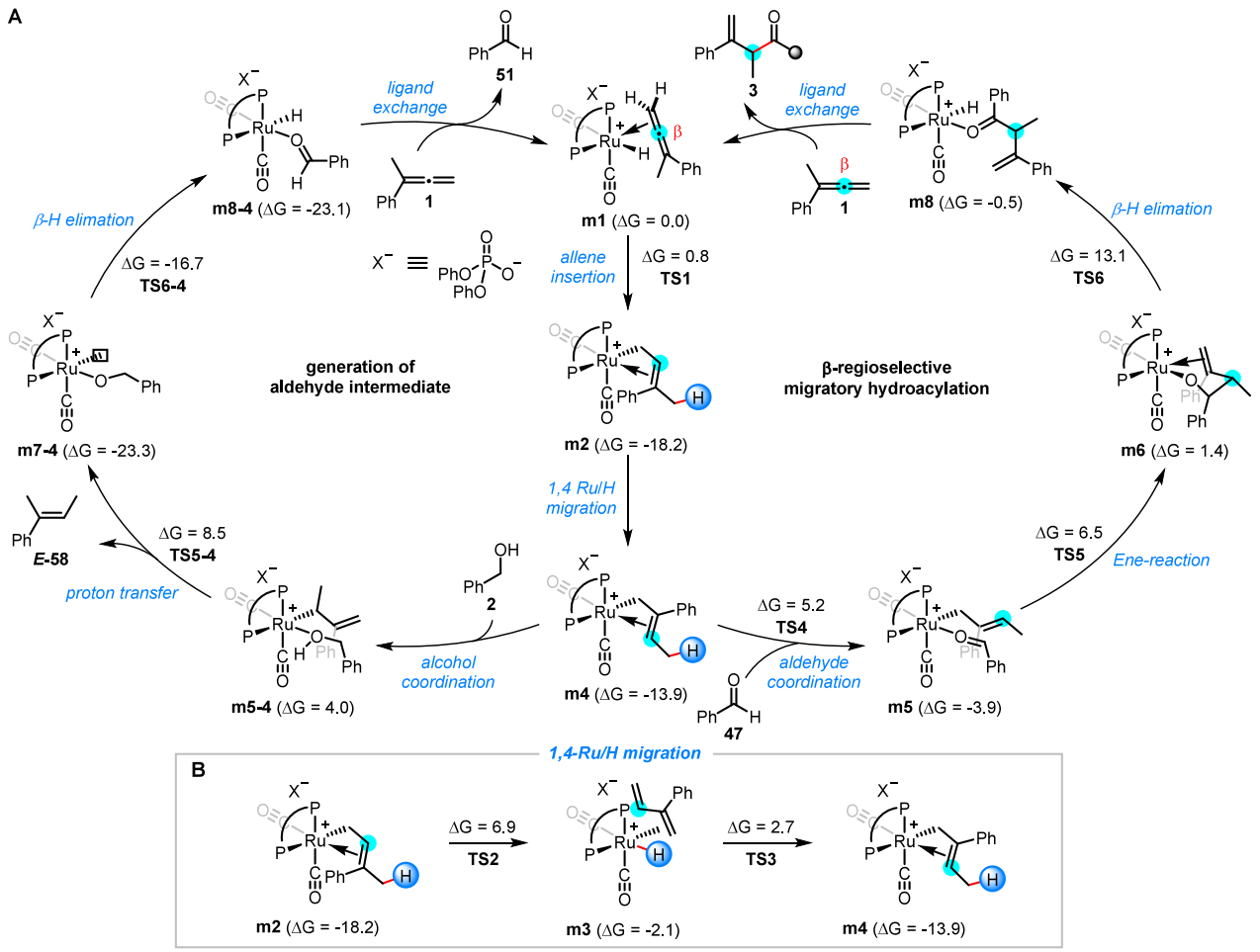


Figure 7. A. Computational result for the mechanism. B. Process of 1,4-Ru/H migration.

2. Desired product **3** was successfully obtained in 81% yield, indicating that Ru complex **57** is an active catalyst (Figure 5H).

Based on the control experiments and mechanistic studies, we proposed the following plausible mechanism of this *β*-regioselective migratory hydroacylation of allene with alcohol (Figure 6A). First, *β*-hydride elimination from an alcohol-coordinated ruthenium complex generates ruthenium hydride **I**. Then, regio- and stereoselective hydroruthenation of allene at its

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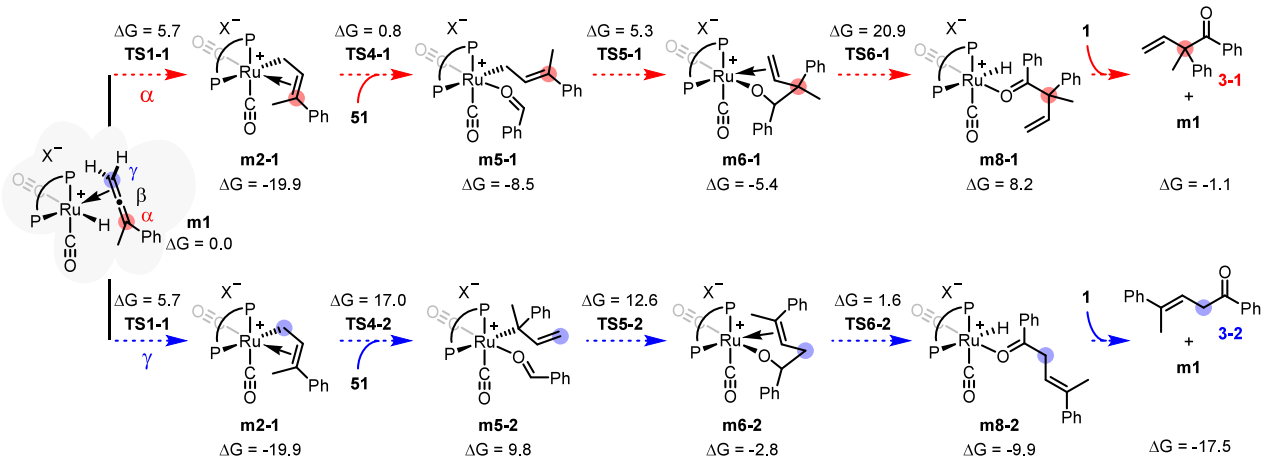


Figure 8. Free energies for the hydroacylation of allene with α- and γ-regioselectivity (ΔG, kcal/mol).

sterically less hindered double bond (γ-position) produces allylic ruthenium species **II**, which can be protonated by alcohol to deliver corresponding alkene product **III** and ruthenium alkoxide **IV**. This was verified by the isolation of alkenes *E*-**53** and *Z*-**53**. The following β-hydride elimination produces aldehyde intermediate **V** and regenerates ruthenium hydride catalyst **I**. Simultaneously and alternatively, allylic ruthenium intermediate **II** could also undergo 1,4-ruthenium/hydrogen migration, leading to allylic ruthenium species **VII**. The following coordination and ene-type allylic addition of in situ-generated aldehyde **V** with **VII** delivers homoallylic ruthenium alkoxide **IX** via Zimmerman−Traxler transition state **VIII**.[56] Finally, β-hydride elimination produces the desired β,γ-enone product and regenerates ruthenium hydride catalyst **I**. The reaction takes place via a process of formal dual C−H functionalization.

The possible process of 1,4-ruthenium/hydrogen migration has been proposed as follows (Figure 6B). Allylic isomerization of allylic ruthenium intermediate **II** generates allylic ruthenium species **IIc**. Then, β-hydride elimination at the ortho-methyl group delivers ruthenium hydride closely coordinated 1,3-diene intermediate **VI**.[49,50] The following fast, formal intramolecular migratory insertion of Ru−H into the sterically less hindered double bond produces resonance-stabilized allylic ruthenium species **VIIa**, which is isomerized to more stable terminal allylic ruthenium intermediate **VII**. This process was confirmed by the detection of various alkene products and the successful hydroacylation of 2-phenyl-1,3-butadiene with benzyl alcohol under the standard conditions.

Next, the proposed catalytic pathway for this β-regioselective migratory hydroacylation has been computed with allene **1** and benzyl alcohol (**2**) using the DFT theory at the M06-L/def2-TZVP/SMD(toluene)//M06-L/6-31G(d, p) (SDD for Ru)/SMD(toluene) level (Figure 7A, details in the Supporting Information). Starting from allene **1**-coordinated Ru−H intermediate **m1**, the insertion of allene occurs via transition state **TS1** with a tiny free-energy barrier of 0.8 kcal/mol. With an exothermic process of 18.2 kcal/mol, allylic ruthenium intermediate **m2** is generated with alkene moiety coordination. Then, two pathways take place, affording the intermediate of aldehyde **47** and β-regioselective migratory hydroacylation product **3**. The 1,4-Ru/H migration occurs via Ru-coordinated 1,3-diene intermediate **m3** generating allylic ruthenium intermediate **m4** via transition states **TS2** and **TS3** with free-energy barriers of 25.1 and 4.8 kcal/mol, respectively (Figure 7B). Then, alkene *E*-**58** can be obtained via protonation with the benzyl alcohol by overcoming an energy barrier of 22.4 kcal/mol. A similar energy barrier of 20.9 kcal/mol is overcome for the formation of *Z*-**58**. At the same time, benzyl alkoxy ruthenium intermediate **m7-4** is formed, which undergoes β-hydride elimination to afford **m6-4** with a free-energy barrier of 6.6 kcal/mol (**TS6-4**). Ligand exchange produces an intermediate of aldehyde **47** and regenerates ruthenium hydride **m1**. As an alternative pathway, the coordination of allylic ruthenium intermediate **m4** with aldehyde **51** instead of the alkene moiety generates intermediate **m5** via metathesis transition state **TS4** with a free-energy barrier of 19.1 kcal/mol. Then, the ene-reaction occurs, leading to homoallylic ruthenium alkoxide intermediate **m6**. The corresponding free-energy barrier for the single step is 20.4 kcal/mol, lower than that of the protonation process. After that, intermediate **m6** undergoes β-hydride elimination, giving carbonyl coordinated complex **m8** with a free-energy barrier of 11.7 kcal/mol. Finally, the ligand exchange between **m8** and the substrate of allene **1** produces final product **3** and regenerates Ru−H catalyst **m1**. The computational studies provide a detailed catalytic pathway and are consistent with our proposed catalytic pathway in Figure 7.

To understand the unique β-regioselectivity of this hydroacylation, we also performed the detailed computation for the α- and γ-regioselective hydroacylation of allene **1** from Ru−H complex **m1** (Figure 8). For the α-regioselective hydroacylation of allene **1** with benzyl alcohol (**2**) toward ketone product **3-1**, the β-hydride elimination step would dominate the process with a free-energy barrier of 40.8 kcal/mol in the global process related to the free energy of **m6-1**. For the γ-regioselective hydroacylation to ketone **3-2**, initial hydroruthenation intermediate **m2-1** will offer a coordination site for aldehyde **51** to intermediate **m5-2** with a free-energy barrier of 36.9 kcal/mol, which would be difficult to pass. In comparison, for β-regioselective migratory hydroacylation, the β-hydride elimination would be the highest step in the whole process, with a free-energy barrier of 31.3 kcal/mol.

CONCLUSIONS

In summary, we have disclosed the first β-regioselective migratory hydroacylation of allenes with readily available alcohols via formal dual C−H functionalization. The reaction occurs efficiently through tandem hydrogen transfer catalysis with an in situ-generated aldehyde as the intermediate. The

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broad scope and good functionality tolerance of this method have been applied to produce versatile β,γ-enone compounds. Moreover, the atom- and step-economical β-regioselective migratory hydroacylation of allene with aldehydes has also been achieved. Unusual site selectivity has been achieved with coupling at the central *sp*-hybridized β-carbon of allenes via a novel mechanism different from traditional metal-catalyzed hydroacylation of allenes with aldehydes. A unique 1,4-metal/hydrogen migration occurs via a 1,3-diene intermediate in this reaction, verified by detailed control experiments and computational studies. We believe that this finding will be insightful for the further design of useful reactions.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acscatal.5c06836.

Materials and methods, experimental procedures, mechanistic studies, DFT calculations, NMR spectra and mass spectrometry data (PDF)

Crystallographic data for compound **57** have been deposited with the Cambridge Crystallographic Data Centre under accession code CCDC 2405767 (CIF)

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Notes

The authors declare no competing financial interest.

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